

Data Preprocessing & Cleaning

Deduplication → quality filtering → PII removal → format standardisation: the data pipeline

9 min

Duration

B3 – Apply

Bloom's

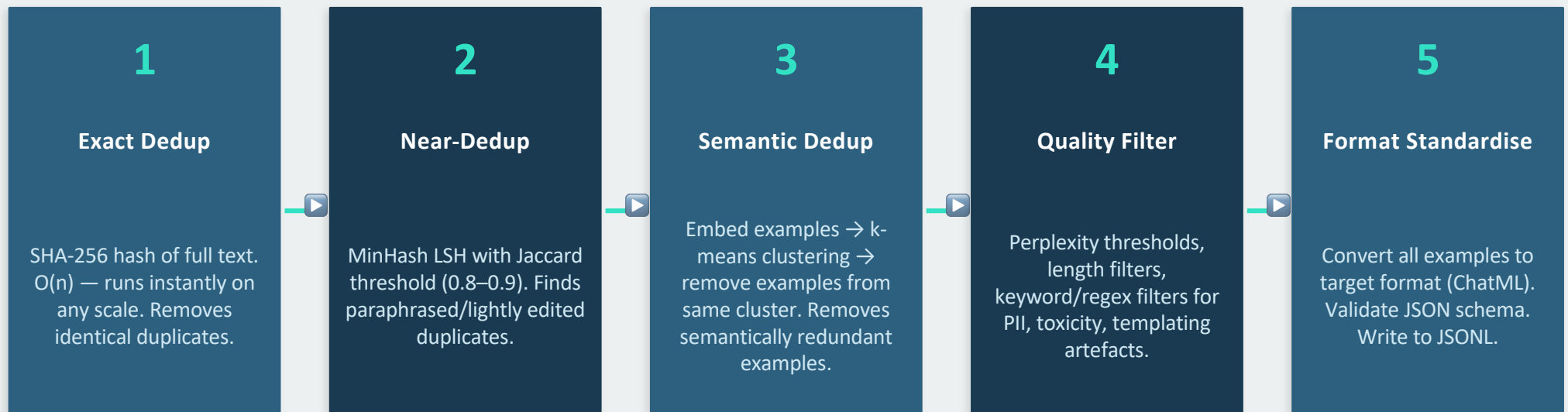
L2 – Intermediate

Level

AI-FT-IN-TH-000016

ID

The Complete Data Cleaning Pipeline



Perplexity-Based Quality Filtering

```
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch, math

# Use a reference model to compute perplexity of each training
example
ref_model_id = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"
ref_tokenizer = AutoTokenizer.from_pretrained(ref_model_id)
ref_model = AutoModelForCausalLM.from_pretrained(ref_model_id,
torch_dtype=torch.bfloat16)

def compute_perplexity(text, max_len=512):
    inputs = ref_tokenizer(text, return_tensors="pt",
truncation=True, max_length=max_len)
    with torch.no_grad():
        loss = ref_model(**inputs, labels=inputs["input_ids"]).loss
    return math.exp(loss.item())

# Filter: remove examples with very low perplexity (trivial) or
very high (incoherent)
filtered = [
    ex for ex in dataset
    if 15 < compute_perplexity(ex["output"]) < 200
]
print(f"Kept {len(filtered)}/{len(dataset)} examples after
perplexity filter")
# Low perplexity (<15): model already knows this well – low
training value
# High perplexity (>200): likely incoherent, toxic, or garbled text
```

Perplexity filtering is one of the most effective quality filters. It identifies both trivially easy examples and incoherent ones — both are bad for fine-tuning.

Quality Filter Catalogue

Filter	What It Removes	Tool/Method
Exact dedup	Identical examples	hashlib SHA-256
Near-dedup	80–90% similar examples	datasketch MinHashLSH
Semantic dedup	Semantically redundant examples	sentence-transformers + sklearn KMeans
Length filter	Too short (<20 words) or too long (>2K)	len(text.split())
Perplexity filter	Trivial or incoherent text	Reference LM loss
PII filter	Emails, phone numbers, SSNs	regex + presidio library
Toxicity filter	Harmful, offensive content	Perspective API / detoxify
Template artefact	Leftover prompts, 'As an AI...'	Custom regex patterns

KEY REFERENCES

- Wenzek, G. et al. (2019). CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data. arXiv.
- Soldaini, L. et al. (2024). Dolmino. arXiv:2402.00159
- presidio library (PII detection). github.com/microsoft/presidio

 **Next: Concept 27: Fine-Tuning Evaluation Metrics (AI-FT-IN-TH-000017)**